scCross: Single-cell multi-omics data integration and cross-modal generation

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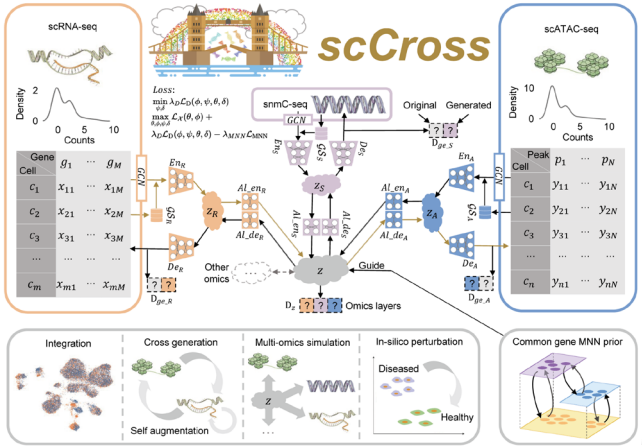
MEAKINS CHRISTIE 50th 1972–2022

Background

Multi-omics datasets present unrivaled opportunities to understand complex biological processes from comprehensive perspectives, but the distinct feature spaces associated with different modalities pose challenges in data integration and downstream analyses. Here we report scCross, an algorithm based on the variational that can integrate the features from different modalities to learn joint cell embeddings. We applied scCross to integrate dual-omics and triple-omics data. scCross is able to map all data modalities to a shared latent space, allowing for various downstream data analytic tasks. scCross outperforms existing methods for data integration and is able to generate single-cell data across modalities. Moreover, scCross can perform multi-omics simulation and infer intra-modal and inter-modal perturbations to manipulate the cellular states, which can be leveraged to discover interventions and therapeutics against various diseases.

Method



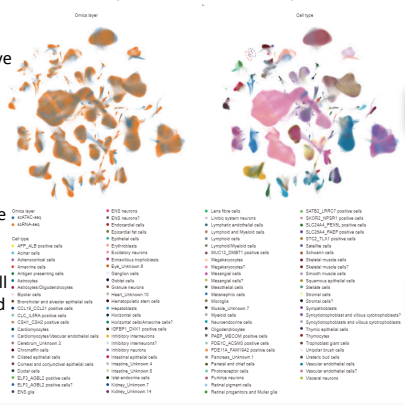
Result

There are two stages of training in scCross. In the first stage, we focus on training the Variational Autoencoder (VAE) for each modality separately. The objective in this stage is to learn the specific bio-information within each modality before integration. We also curating gene sets from Gene Ontology and Pathway terms to enhance the training.

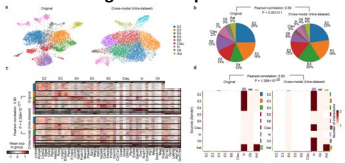
In the second stage of training, we build upon the trained VAE models from the first stage and introduce bi-directional aligners to align the modalities together in a common latent space u. The discriminators also play a role in this stage. Mutual Nearest Neighbors (MNN) correction based on the common genes in all modalities is utilized to generate a prior to guide the alignment between modalities to match the cells with similar cellular states.

Large scale single cell data integration facing challenges like sheer volume of data, extensive heterogeneity, low coverage per cell and unbalanced cell type compositions. Besides these challenges, minority cell type populations are easily be over-integrated to majority cell type populations, which is still yet be accomplished and hard to be reflected by normal overall estimations. We do well in all those aspects.

Integration performance on Human Cell Atlas (Over four million cells)

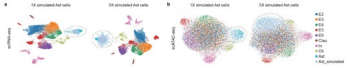


Cross generation performance



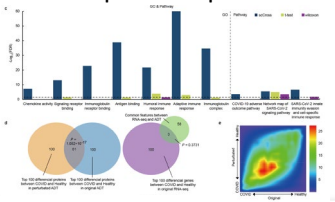
These four plots show the similarity between cross generated scRNA-seq data from scATAC-seq data and the original scRNA-seq data. The data used for generation is totally unlearned by our model.

Multi-omics simulation performance



We simulated both scRNA-seq and scATAC-seq for Ast cells for 1 and 5 times.

In-silico perturbation performance



The upper plot shows the pathways finding based on DEgenes gain from intra modality perturbation.

The bottom two plots below show the cross generated data maintains the perturbation we exert to the other modality we use to perform the generation.

Conclusions

Based on VAE-GAN deep generative model, common gene MNN prior and gene set information, scCross is designed to perform integration, cross-modal generation, multi-omics simulation and in-silico perturbation among all kinds of single cell muti-omics data. Whatever kinds of single cell omics data researchers own (matched or unmatched, two modalities or more), scCross can help them study in a solid view.

Acknowledge

McGill University Health Centre Research Institute

MEAKINS CHRISTIE

CIHR IRSC

Fonds de recherche Santé Québec

THREE LAKES FOUNDATION

NSERC CRSNG